MIL-A-38329(USAF)
AMENDMENT 5
16 OCTOBER 1987
SUPERSEDING
AMENDMENT 4
01 APRIL 1982

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SET
AN/ASN-55

This amendment forms a part of MIL-A-38329(USAF), dated 3 August 1964, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Under SPECIFICATION, add: "FEDERAL" and the following: "PPP-B-601 Boxes, Wood, Cleated Plywood" and "PPP-B-636 Box Shipping, Fiberboard."

Under "SPECIFICATION, MILITARY" delete "MIL-M-26512" and "MIL-R-26667" and add the following:

"MIL-P-116	Preservation, Methods Of
MIL-I-18309	Integrated Electronic Central AN/ASQ-19()"

Add the followong boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Lower right-hand corner, delete "FSC 6610" and add: "FSC 6605".

PAGE 2

Under "STANDARDS, MILITARY" delete " MIL-STD-831" and add the following:

"MIL-STD-471	Maintainability Demonstration
MIL-STD-2073/1A	DOD Material Procedure For Development And Application Of Packaging Requirement"

* Under Drawing, Air Force, delete "64J1546 and title."

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* Add: "2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawing, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation: Drawing AIR FORCE 64J1546 Schematic Wiring Dragram-Reference Set Attitude Heading AN/ASN-55."

* 2.2 Delete and add:

"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

* 3.1 Delete and add:

"3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4)."

3.2, a Delete "ADK-182/A24G-1A" and add: "ADK-182A/A24G-1A"

PAGE 4

"3.3f Delete "AN/ASQ-88" and add: "AN/ASQ-19()"

3.4.1 Delete paragraph and add:

"3.4.1 Reliability. The reference set shall have a minimum acceptable mean-time-between-failure (MTBF) of 250 hours based on a discrimination ratio of 1.5 and a confidence level of 90 percent.

3.4.1.1 Line 3 delete "MIL-R-26667" and add: "MIL-STD-781".

3.4.2 Line 3 delete "indicator" and add: "reference set".

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3.5.2 Line 5 and 6, delete "ADK-182/A24G-1A" and add: "ADK-182A/A24G-1A."

PAGE 8

Table I: Under "Excitation" for output signal No. 6, add "dc" after "28V".

Table II: Under "Signal Output" delete "DC drift" and add: "DG drift".

Table II: Under "Dynamic Error" for "DG drift" delete "2°" and add: "6°/hr".

PAGE 9

3.7 Line 3 delete "ADK-182/A24G-1A" and add: "ADK-182A/A24G-1A."

3.8 Line 5 delete "cycle".

3.9 Line 1 delete "ADK-182/A24G-1A" and add: "ADK-182A/A24G-1A."

Add the following new paragraph to Section 3:

"3.13 System compatibility. The system components shall be considered compatible after satisfactory completion of the tests specified in 4.4.3.

* "3.14 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

* Add "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

* 4.2, a, b Delete and add:

"4.2 Classification of inspections. The requirements specified herein are classified as follows:

a. First article inspection (See 4.4.3).

b. Quality conformance inspection (see 4.5)."

* 4.3 Delete and add:

"4.3 Inspection condition. Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.3.2 of this specification."

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PAGE 11

4.3.1 Delete the third and fourth sentences and add:"The synchro transmitter shall be connected to J502-a through -f. The type MD-1 gyroscope will not be required when the synchro transmitters are used."

4.3.3,d Line 2, Delete "$\pm 10^{o}$ per minute" and add: "$\pm 14^{o}$ per minute".

PAGE 12

4.3.4.1, R5 After "23.5V" add "ac".

PAGE 13

Figure 1: Delete and add the attached Figure 1.

PAGE 15

* 4.4 Delete and add: "4.4 First article inspection."

4.4.1.1 Lines 2 and 3, Delete "shall be selected at random from the first 50 produced on the contract or order and" and add: "representative of the production equipment on the contract or order shall be."

* Delete "4.4.2".

* 4.4.3 Delete and add:

"4.4.3 First article inspections. The first article inspections shall consist of all the tests specified under 4.6".

* 4.5 Delete and add:

"4.5 Quality conformance inspections. Quality conformance inspection shall consist of the sampling tests.

PAGE 19

4.5.3 Line 4 between "4.6.15" and "for" insert "except 4.6.15.3".

PAGE 22

4.6.4 Line 4 delete "120°" and add: "170°".

4.6.5 Line 2 delete "120°" and add: "170°". At end of third sentence, add "$\pm 2^{o}$ within 5 seconds". Line 7, delete "120°" and add "170°". Last sentence, add "$\pm 2^{o}$ within 5 seconds".

4.6.6 Line 3 delete "120°" and add: "170°".

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PAGE 23

4.6.11 Line 5 delete "attitude indicator" and add: "servoed repeater".

PAGE 27

4.6.15.2.5 Line 1 delete "COMP-DG-SLAVED-SYNC" and add: "mode select".

4.6.15.2.7 Line 3 between "climb" and "and", insert "(ccw)".

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4.6.15.2.12.1 Line 1 delete "COMP-DG-SLAVED-SYNC" and add: "mode select".

PAGE 29

4.6.15.2.15 Line 5 delete "6.0°" and add: "2.0° (8°/hour)".

PAGE 30

Add: "4.6.15.2.18 Primary synchronization. With the azimuth heading shown on the attitude indicator synchronized to 0° heading set in on the compass transmitter simulator, 170° heading shall be set in on the compass transmitter simulator. The PRIM-STBY switch on the compass controller shall be operated to the PRIM position. The azimuth heading shown on the attitude indicator shall sync to within 2° of the 170° heading set in on the compass transmitter simulator within 5 seconds, maximum, and the sync indicator shall approach a null position."

Add: "4.6.15.2.19 Standby synchronization. With the azimuth heading shown on the attitude indicator synchronized to the 170° heading set in on the compass transmitter simulator, a 0° heading shall set in on the compass transmitter simulator. The PRIM-STBY switch on the compass controller shall be operated to the STBY position. The azimuth heading shown on the attitude indicator shall sync to within 2° of the 0° heading set in on the compass transmitter simulator within 5 seconds, maximum, and the sync indicator shall approach a null position."

* Delete "4.6.15.5", "a", "b", "c", and "d" in its entirety.

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* Delete 4.6.15.6 in its entirety.

4.6.16 Delete paragraph and add:

"4.6.16 Maintainability demonstration. A maintability demonstration test shall be conducted in accordance with MIL-STD-471 on a minimum of three selected repairs. The tasks shall be selected in advance of timing the tasks and shall be approved by the contracting activity."

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Add new paragraph as follows:

"4.7 Inspection of packaging. Test of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

* Delete Section 5 in its entirety and add:

" 5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073/1A, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Attitude Heading Reference Set shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Attitude Heading Reference Set shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each attitude heading reference set shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each attitude heading reference set shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness in G's established for the attitude heading reference set when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each attitude heading reference set shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of attitude heading reference set be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Attitude heading reference set packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Attitude heading reference set packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

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5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged attitude heading reference set shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

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* Add: "6.4 First article. When a first article inspection is required sample units shall be inspected as specified (see 3.1, 4.2 and 4.3).

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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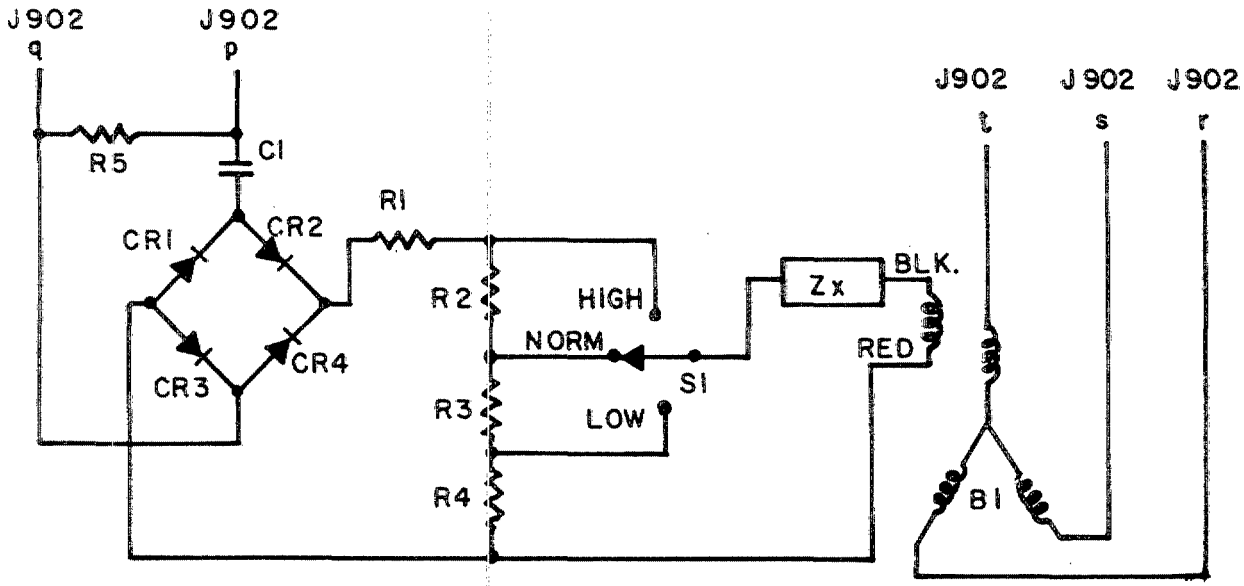


FIGURE 1. Flux valve signal source.

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